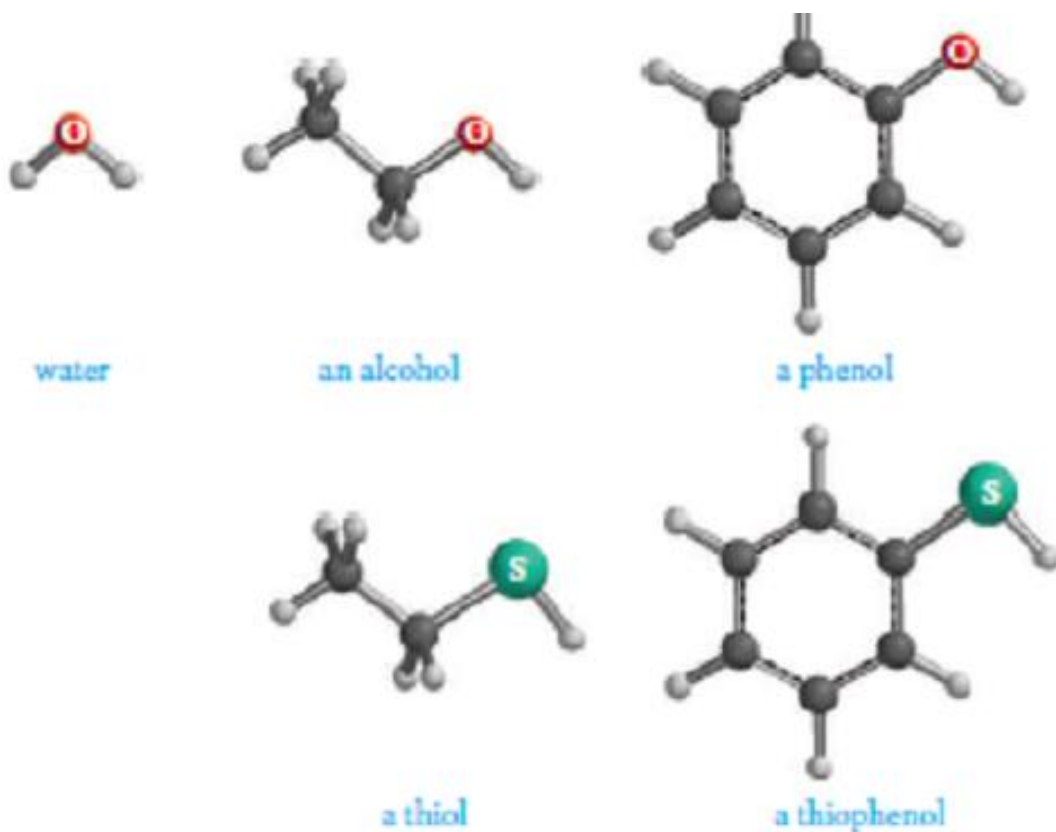
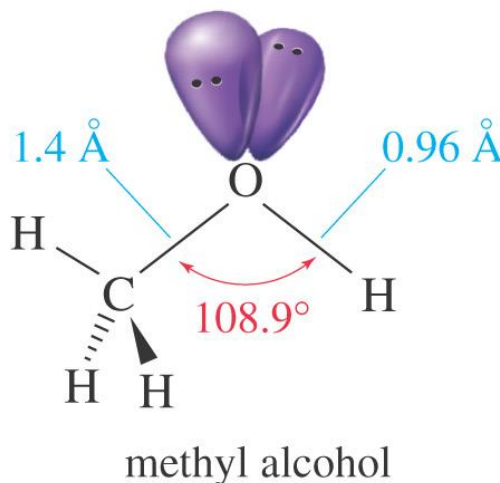
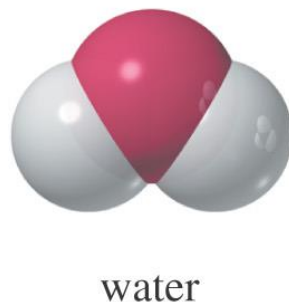
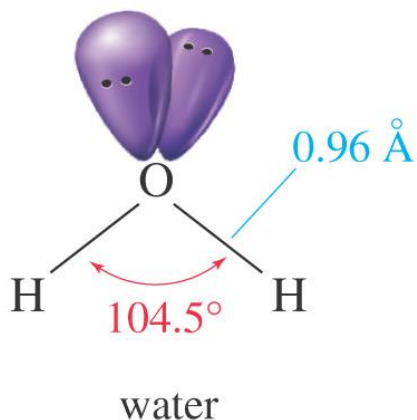


Alcohols, phenols, and Thiols

Alcohols have the general formula $R-OH$ and are characterized by the presence of a **hydroxyl group**, $-OH$. They are structurally similar to water, but with one of the hydrogens replaced by an alkyl group. **Phenols** have a hydroxyl group attached directly to an aromatic ring. **Thiols** and thiophenols are similar to alcohols and phenols, except the oxygen is replaced by sulfur.



7.1 Nomenclature of Alcohols

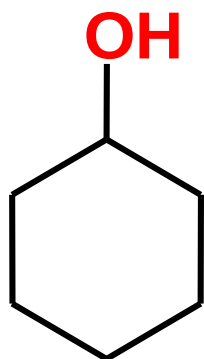


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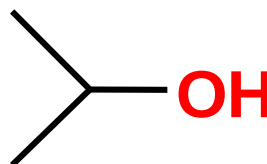
- Hydroxyl (-OH) functional group
- Oxygen is sp^3 hybridized.

7.1 Nomenclature of Alcohols

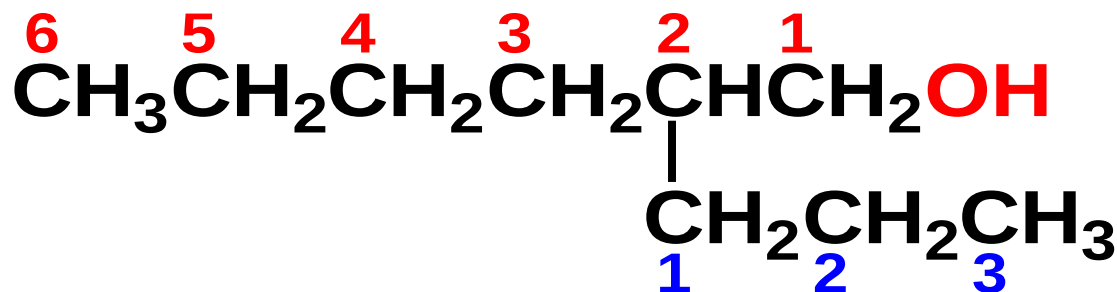
- Common names are the name of alkyl group followed by the word “alcohol”



cyclohexyl
alcohol



isopropyl
alcohol



2-propyl-1-hexanol

Notice that
numbering starts
at alcohol group



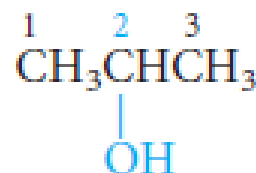
methanol
(methyl alcohol)



ethanol
(ethyl alcohol)



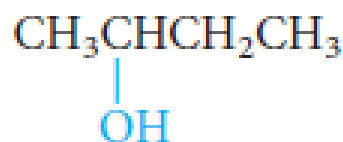
1-propanol
(*n*-propyl alcohol)



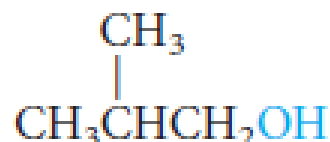
2-propanol
(isopropyl alcohol)



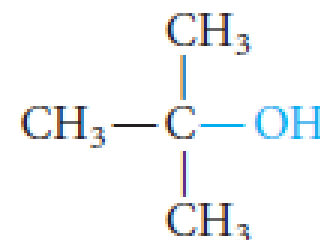
1-butanol
(*n*-butyl alcohol)



2-butanol
(*sec*-butyl alcohol)



2-methyl-1-propanol
(isobutyl alcohol)



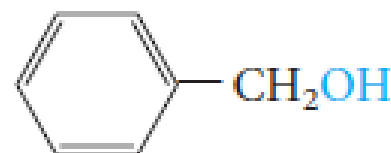
2-methyl-2-propanol
(*tert*-butyl alcohol)



2-propen-1-ol
(allyl alcohol)



cyclohexanol
(cyclohexyl alcohol)

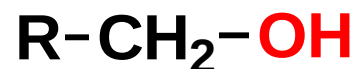


phenylmethanol
(benzyl alcohol)

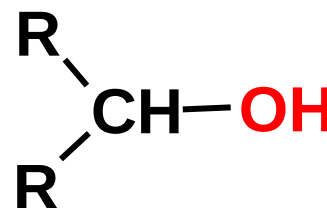
7.2 Classification of Alcohols

- Compounds in which a hydrogen is replaced by an **OH** group.
- We distinguish three types:

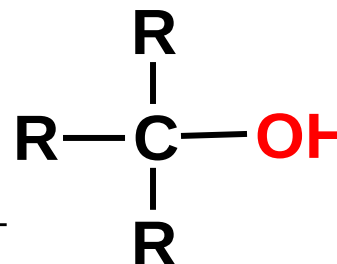
- Primary alcohols



- Secondary alcohols

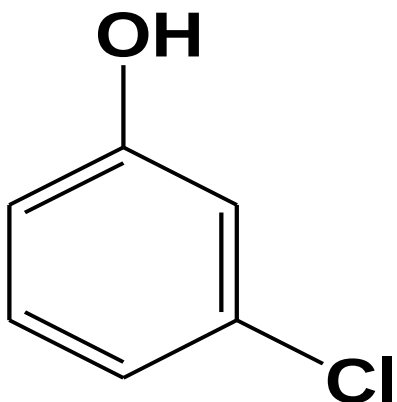


- Tertiary alcohols

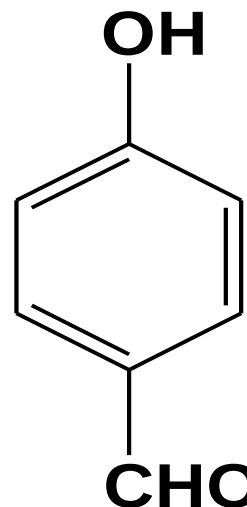


7.3 Nomenclature of Phenols

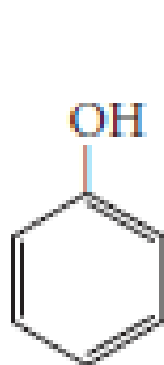
- OH group is assumed to be on carbon 1.
- For common names of disubstituted phenols, use *ortho*- for 1,2; *meta*- for 1,3; and *para*- for 1,4.
- **OH** is named as hydroxy.



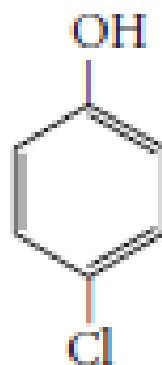
meta-chlorophenol
3-chlorophenol



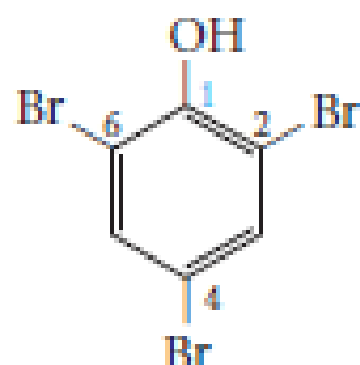
4-hydroxybenzaldehyde



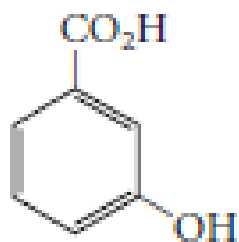
phenol



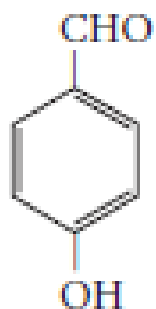
p-chlorophenol



2,4,6-tribromophenol



m-hydroxy
benzoic acid



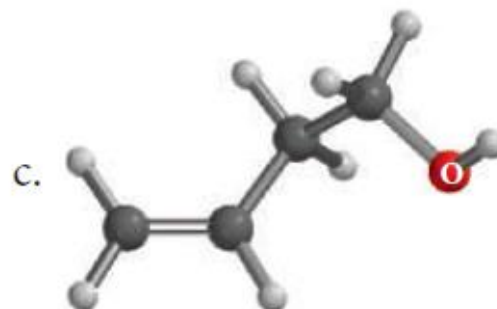
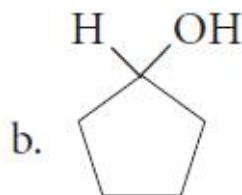
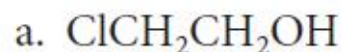
p-hydroxybenzaldehyde

but



p-nitrophenol
(*not p*-hydroxynitrobenzene)

PROBLEM 7.1 Name these alcohols by the IUPAC system:



PROBLEM 7.2 Write a structural formula for

a. 3-methylcyclohexanol

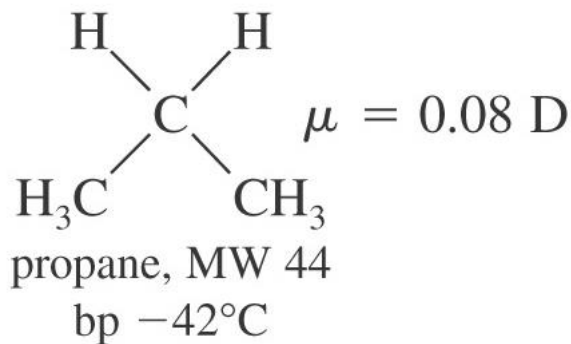
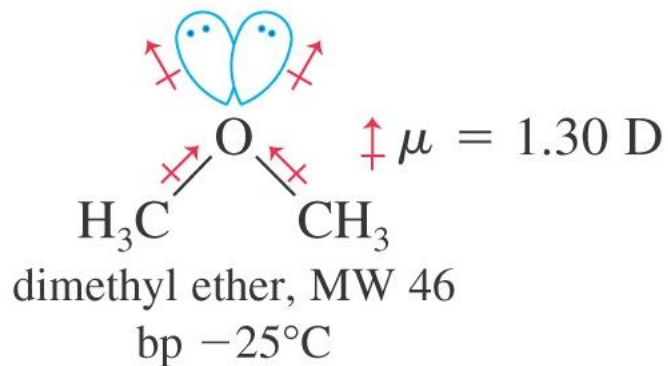
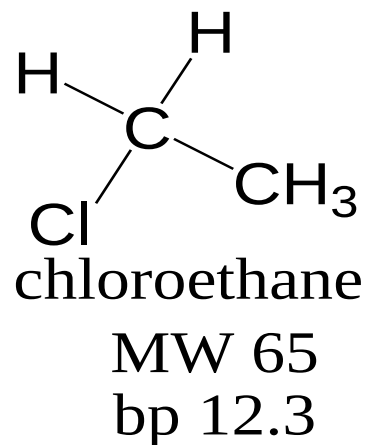
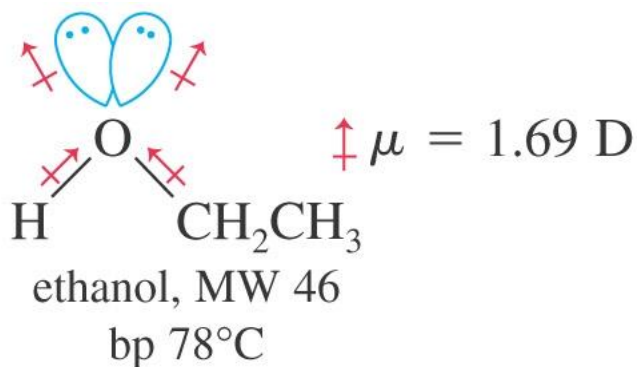
b. 2-phenylpropanol

c. 3-pentyn-2-ol

PROBLEM 7.4 Write the structure for

- pentachlorophenol (an insecticide for termite control, and a fungicide)
- m*-isopropylphenol
- o*-hydroxyacetophenone (for the structure of acetophenone, see Sec. 4.6)
- 3-hydroxy-5-nitrobenzaldehyde

7.4 Hydrogen Bonding in Alcohols and Phenols



Hydrogen Bonding in Alcohols and Phenols

- The boiling points (bp's) of alcohols are much higher than those of ethers or hydrocarbons with similar molecular weights.
- Unusually high boiling points due to hydrogen bonding between molecules.
- A positively polarized —OH hydrogen atom from one molecule is attracted to a lone pair of electrons on a negatively polarized oxygen atom of another molecule
- This produces a force that holds the two molecules together

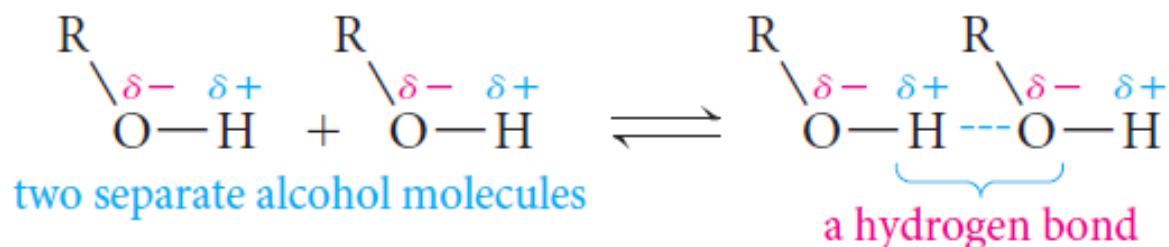


TABLE 7.1 BOILING POINT AND WATER SOLUBILITY OF SOME ALCOHOLS

Name	Formula	bp, °C	Solubility in H ₂ O g/100 g at 20°C
methanol	CH ₃ OH	65	completely miscible
ethanol	CH ₃ CH ₂ OH	78.5	completely miscible
1-propanol	CH ₃ CH ₂ CH ₂ OH	97	completely miscible
1-butanol	CH ₃ CH ₂ CH ₂ CH ₂ OH	117.7	7.9
1-pentanol	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ OH	137.9	2.7
1-hexanol	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ OH	155.8	0.59

Small alcohols are miscible in water, but solubility decreases as the size of the alkyl group increases.

7.7 The Acidity of Alcohols and Phenols

- Alcohols and Phenols are Weak Brønsted Acids
- Can transfer a proton to water to a very small extent
- Produces H_3O^+ and an **alkoxide ion, RO^-** , or a **phenoxide ion, ArO^-**

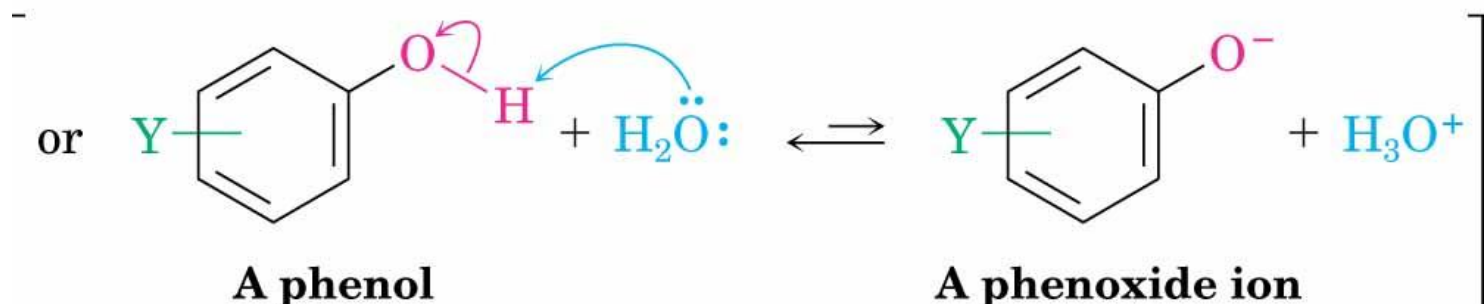
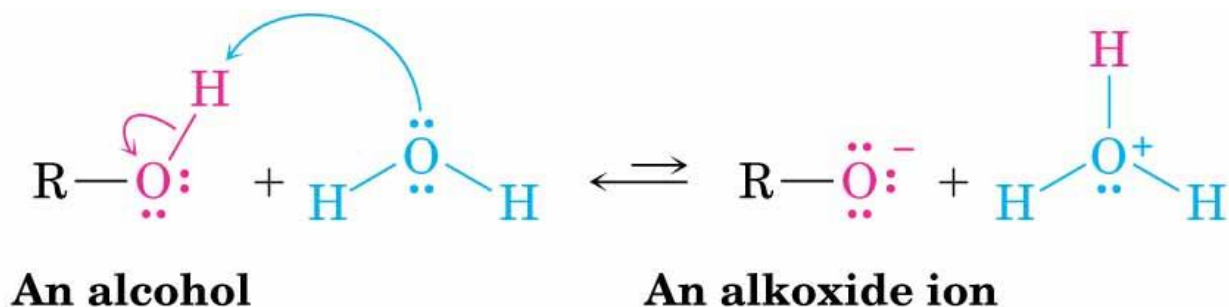
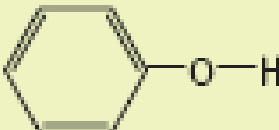
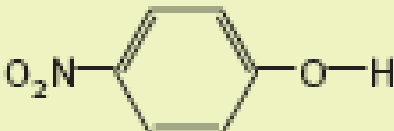
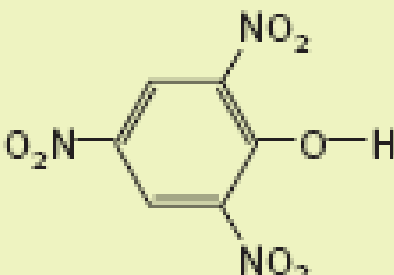


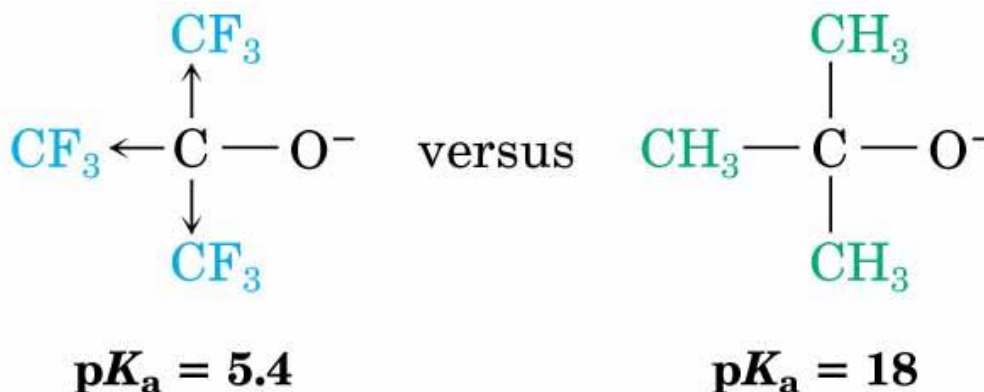
Table 7.2  pK_a 's of Selected Alcohols and Phenols in Aqueous Aolution

Name	Formula	pK_a
water	HO—H	15.7
methanol	CH ₃ O—H	15.5
ethanol	CH ₃ CH ₂ O—H	15.9
<i>t</i> -butyl alcohol	(CH ₃) ₃ CO—H	18
2,2,2-trifluoroethanol	CF ₃ CH ₂ O—H	12.4
phenol		10.0
<i>p</i> -nitrophenol		7.2
picric acid		0.25

Relative Acidities of Alcohols

- Simple alcohols are about as acidic as water.
- Alkyl groups make an alcohol a weaker acid.
- Electron-withdrawing groups make an alcohol a stronger acid by stabilizing the conjugate base (alkoxide)

Electron-withdrawing groups stabilize alkoxide and lower pK_a

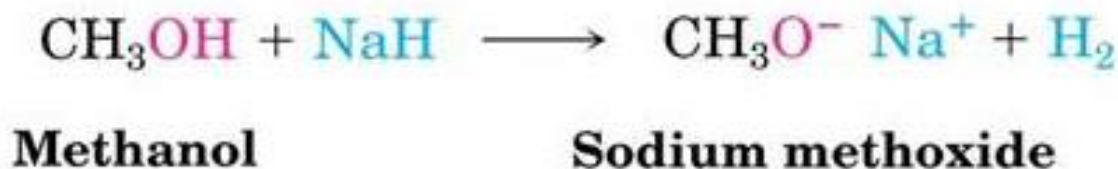
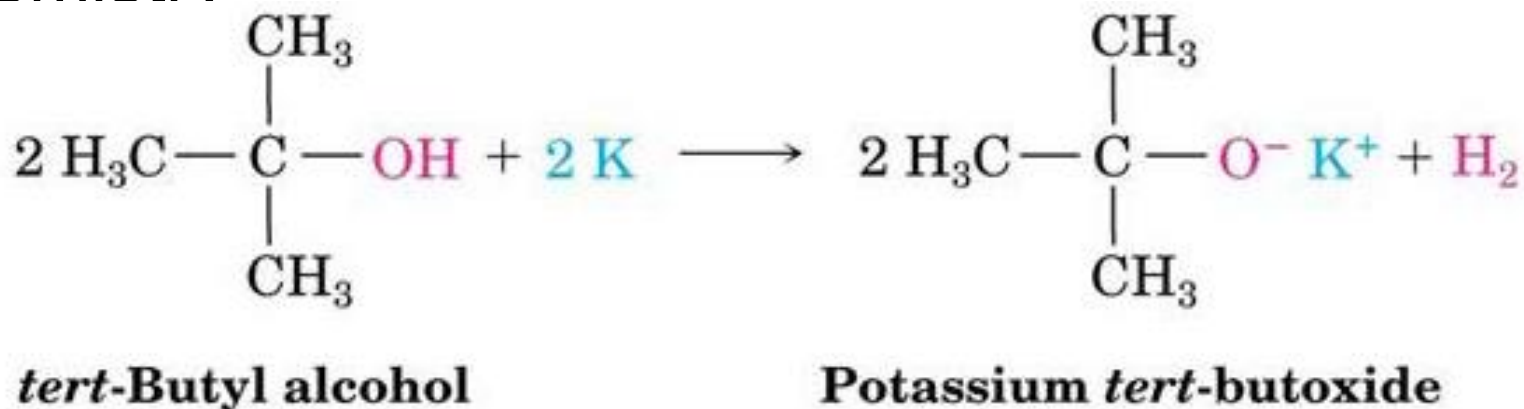


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Inductive Effects

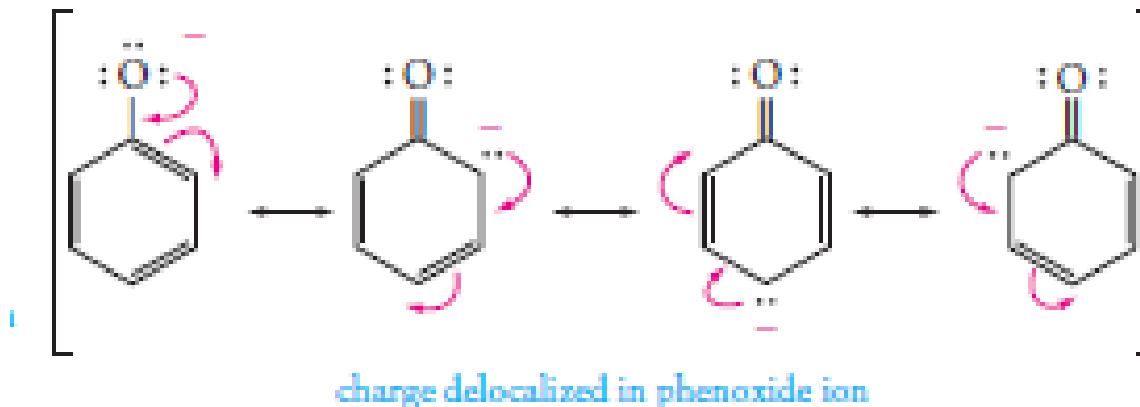
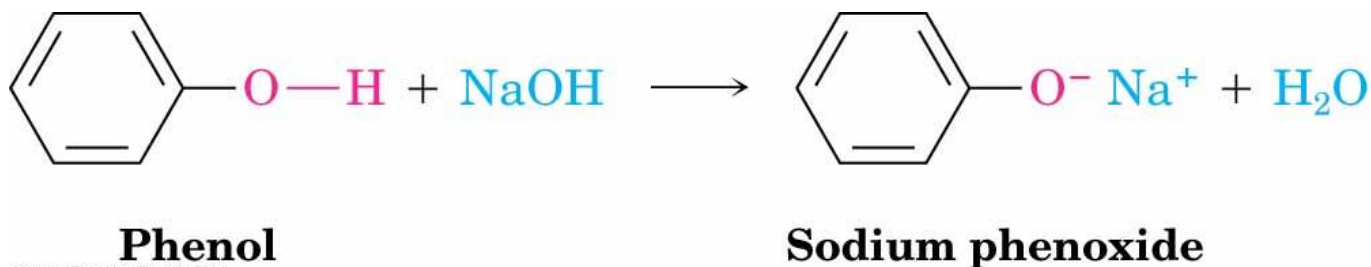
Generating Alkoxides from Alcohols

- Alcohols are weak acids – requires a strong base to form an alkoxide such as NaH, or metal (K, Na)
- Alkoxides are bases used as reagents in organic chemistry



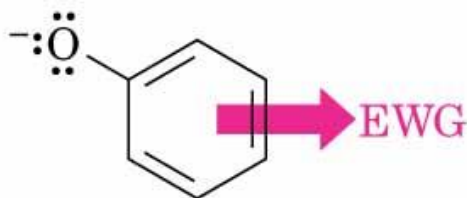
Phenol Acidity

- Phenols ($\text{pK}_a \sim 10$) are much more acidic than alcohols ($\text{pK}_a \sim 16$) due to resonance stabilization of the phenoxide ion
- Phenols react with NaOH solutions (but alcohols do not), forming soluble salts that are soluble in dilute aqueous



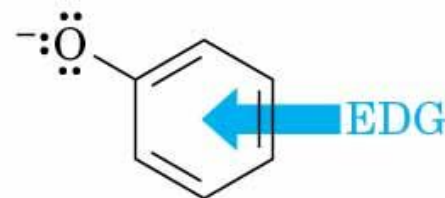
Substituted Phenols

- Can be more or less acidic than phenol itself
- An electron-withdrawing substituent makes a phenol more acidic by delocalizing the negative charge
- Phenols with an electron-donating substituent are less acidic because these substituents concentrate the charge



Electron-withdrawing groups (EWG)
stabilize phenoxide anion, resulting
in increased phenol acidity

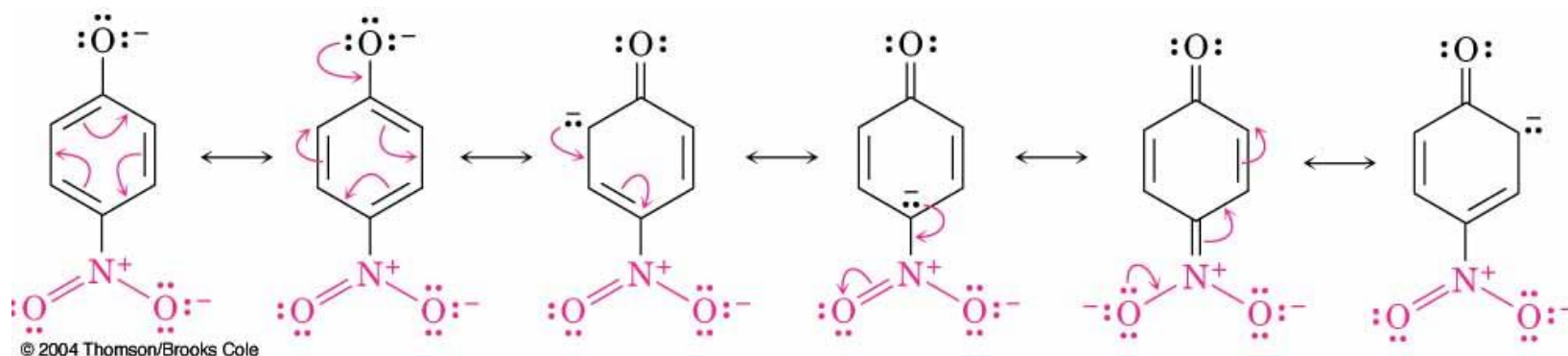
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Electron-donating groups (EDG)
destabilize phenoxide anion,
resulting in decreased phenol acidity

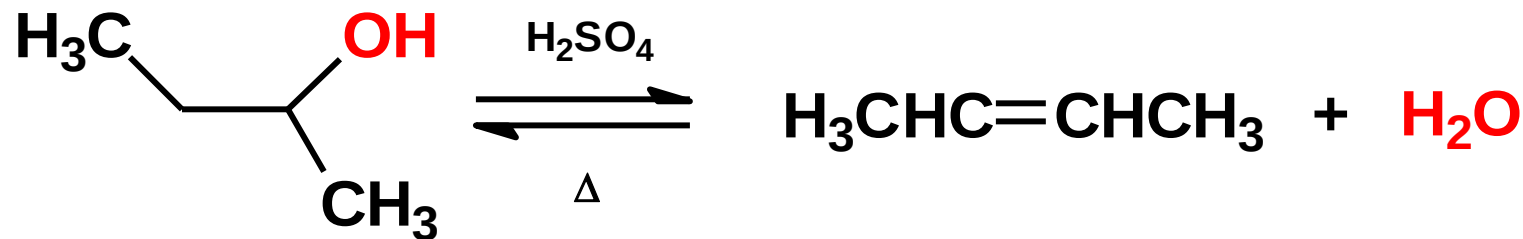
Nitro-Phenols

- Phenols with nitro groups at the ortho and para positions are much stronger acids
- The pKa of 2,4,6-trinitrophenol is 0.6, a very strong acid



7.8 Dehydration of Alcohols to Alkenes

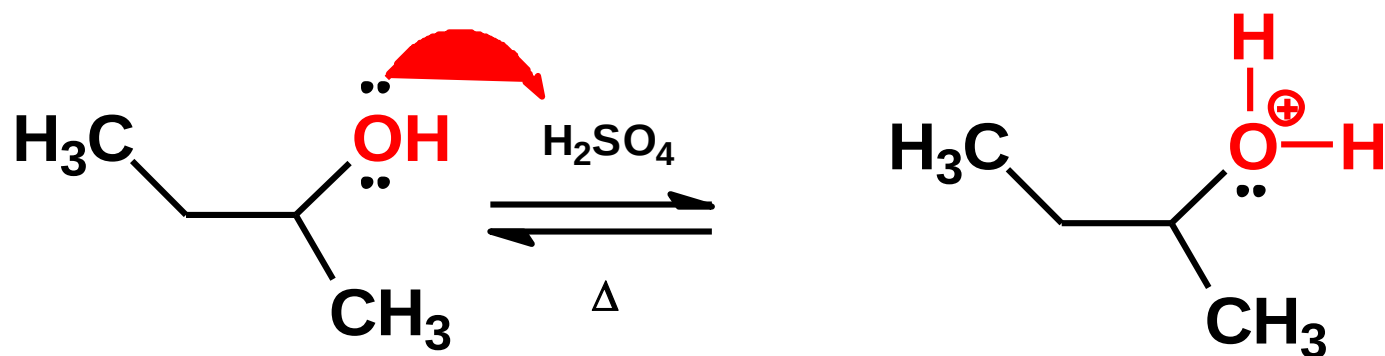
- Dehydration (elimination) of water is commonly obtained using sulfuric acid (H_2SO_4) as a catalyst.



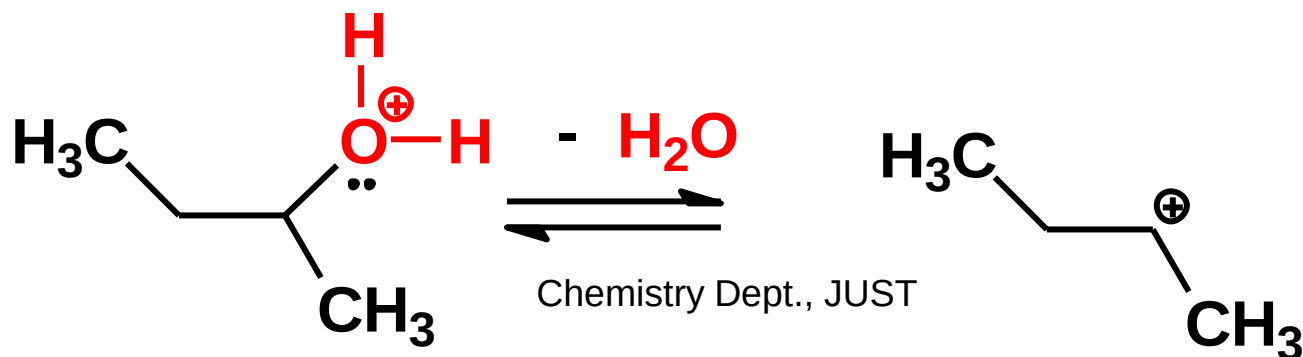
➤ The acid is mandatory to convert the poor leaving group OH^- into a good leaving group H_2O .

E1 Dehydration

- First step is protonation of the hydroxyl group.

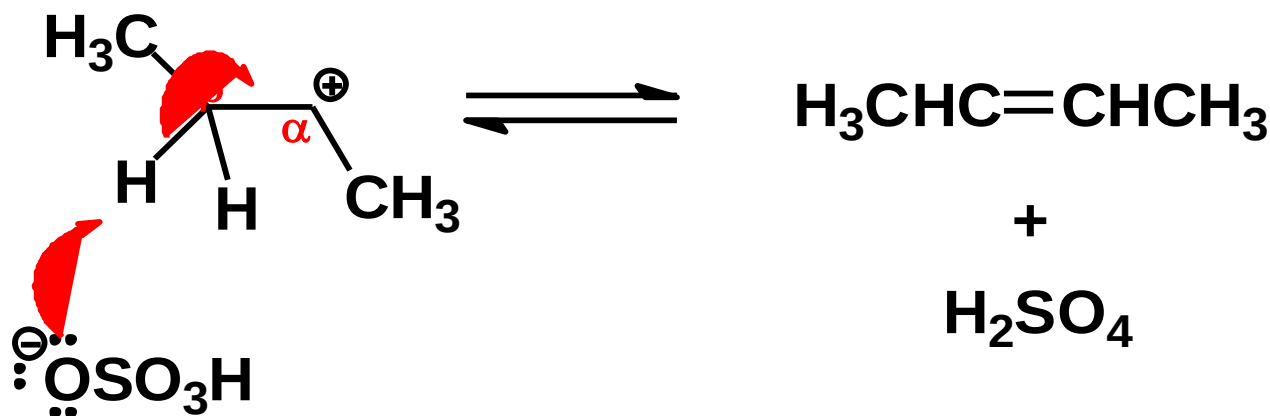


➤ Loss of water leads to a carbocation.



Dehydration

- Second, a base removes a proton β to the carbocation center.

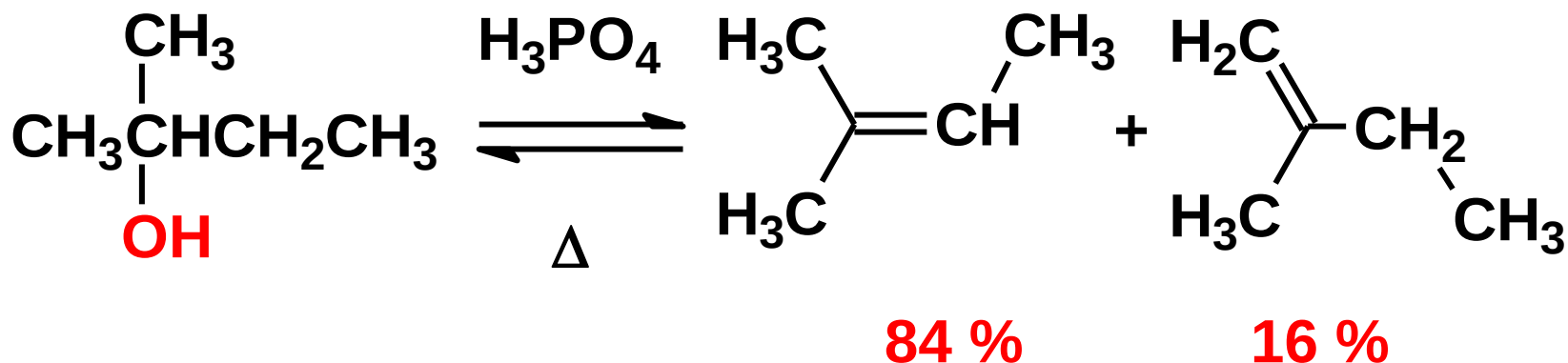


➤ Notice that this reaction is an E1 reaction.

- Rate-determining step is the formation of the carbocation.

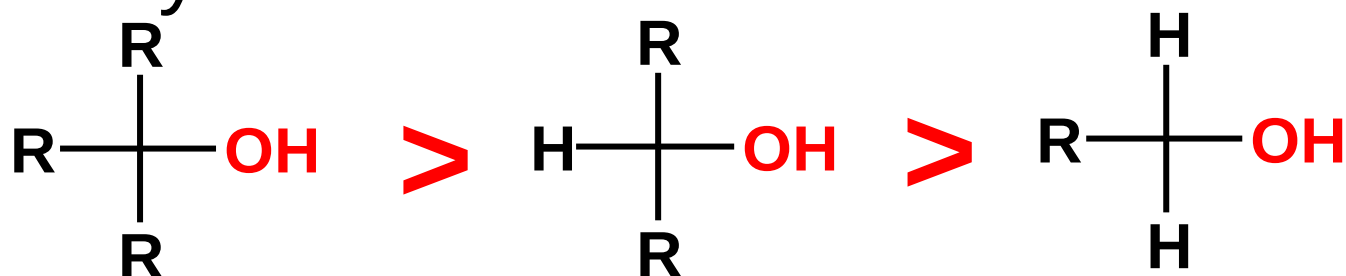
Dehydration

- In case we have a choice between several β -hydrogens, the most stable alkene usually predominates

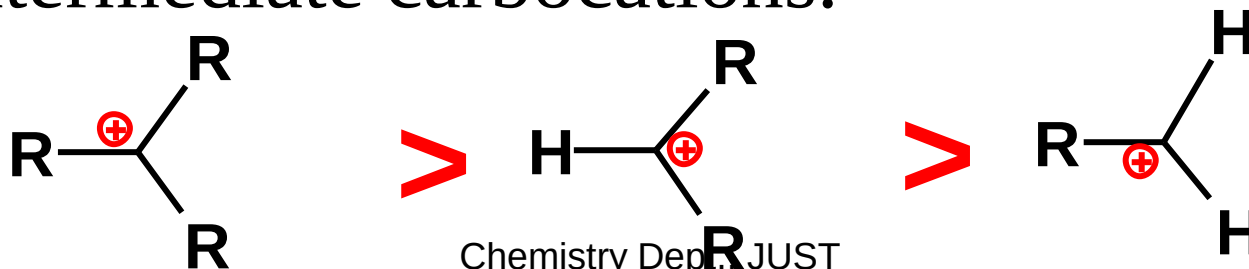


Dehydration

- As a result of the E1 mechanism, the ease of dehydration follows the order:

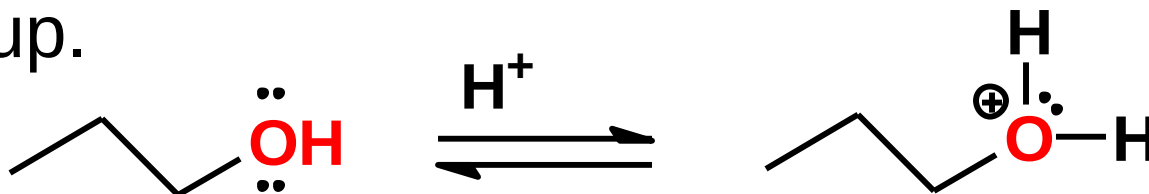


- That directly reflects the stability of the intermediate carbocations.

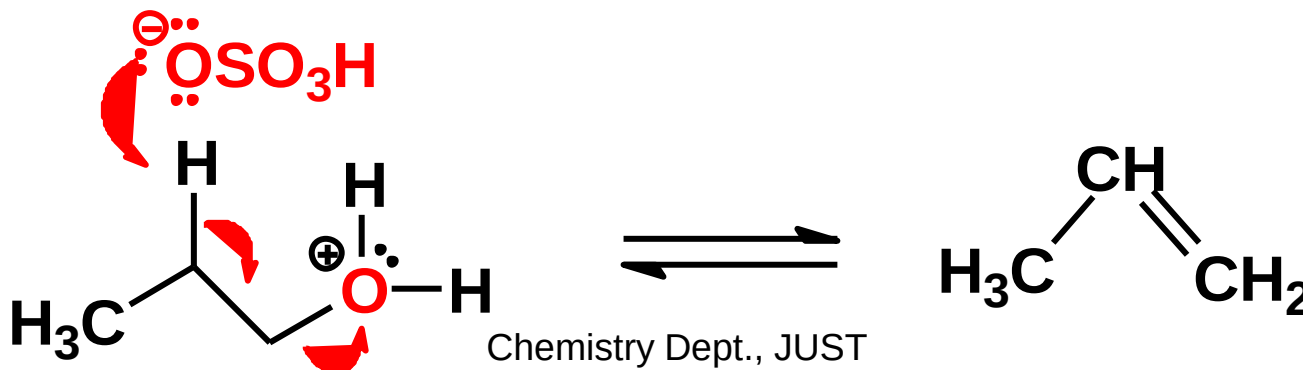


Dehydration

- Primary alcohols undergo dehydration by an E2 pathway.
- First, however, we generate the good leaving group.



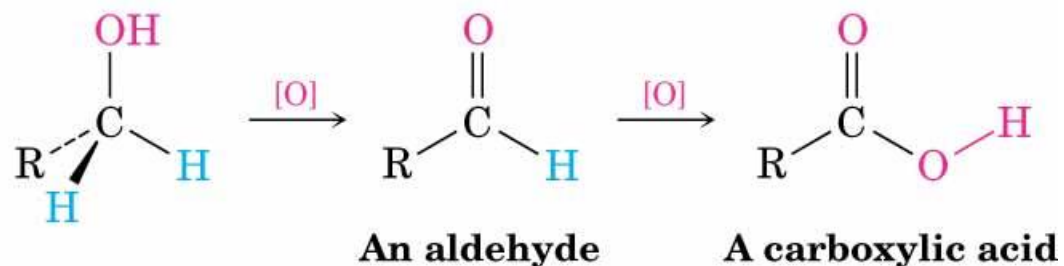
- The subsequent steps, removal of water and deprotonation, take place simultaneously.



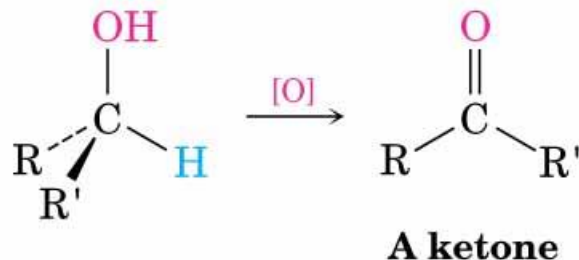
7.12 Oxidation of Alcohols to Aldehydes, Ketones, and Carboxylic Acids

- Can be accomplished by inorganic reagents, such as KMnO_4 , CrO_3 , and $\text{Na}_2\text{Cr}_2\text{O}_7$ or by more selective, expensive reagents

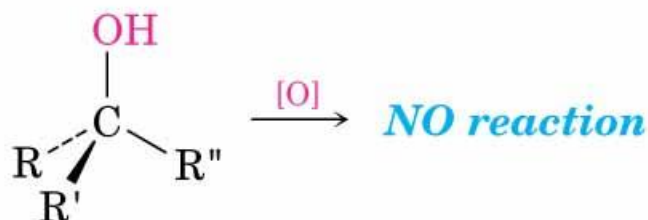
Primary alcohol



Secondary alcohol

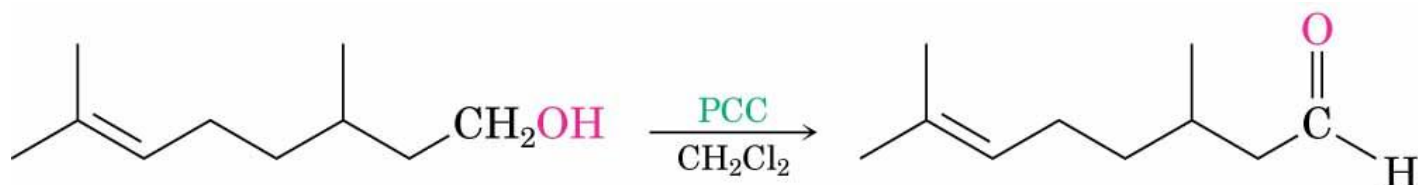


Tertiary alcohol



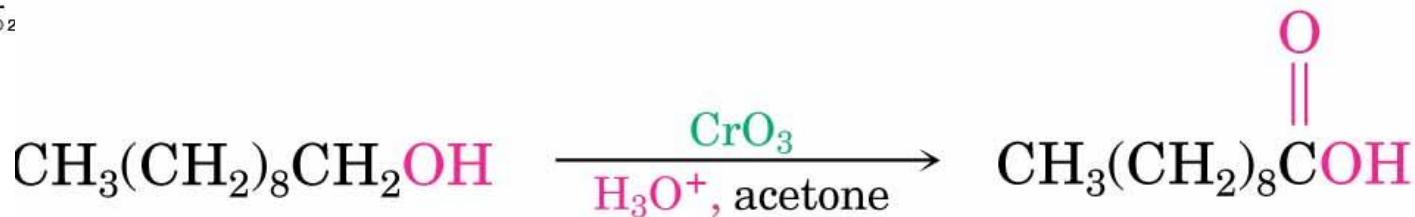
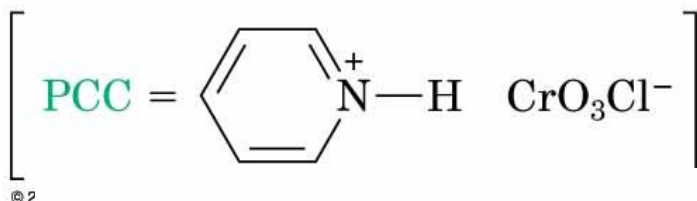
Oxidation of Primary Alcohols

- To aldehyde: pyridinium chlorochromate (PCC, $\text{C}_5\text{H}_6\text{NCrO}_3\text{Cl}$) in dichloromethane
- Other reagents produce carboxylic acids



Citronellol (from rose oil)

Citronellal (82%)



1-Decanol

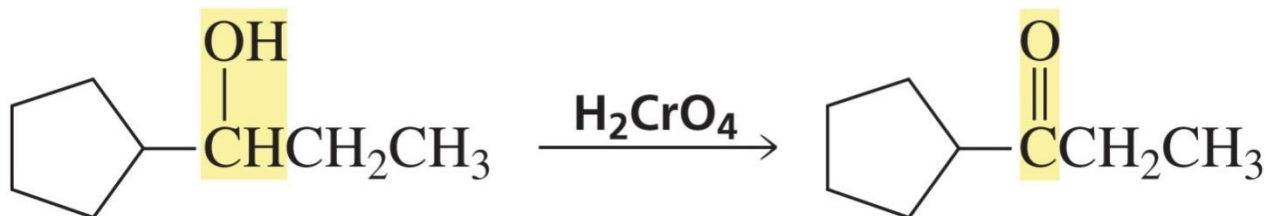
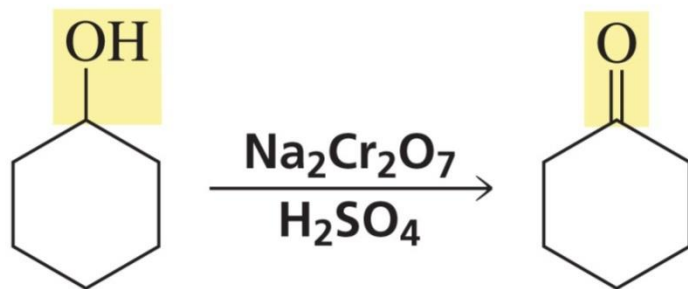
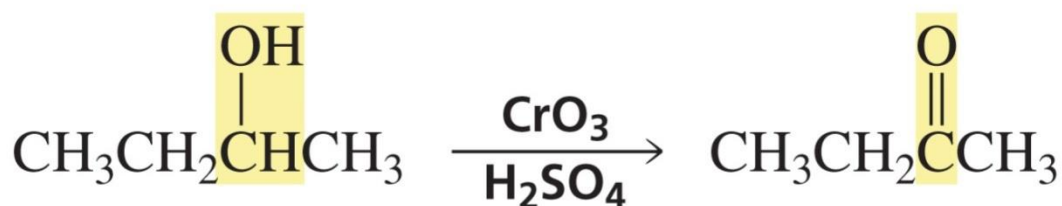
Decanoic acid (93%)

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Oxidation of Secondary Alcohols

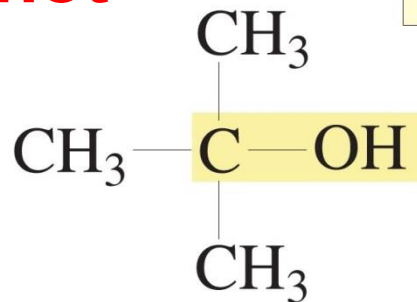
secondary alcohols

ketones



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A tertiary alcohol cannot be oxidized



cannot be oxidized to a carbonyl group

a tertiary alcohol

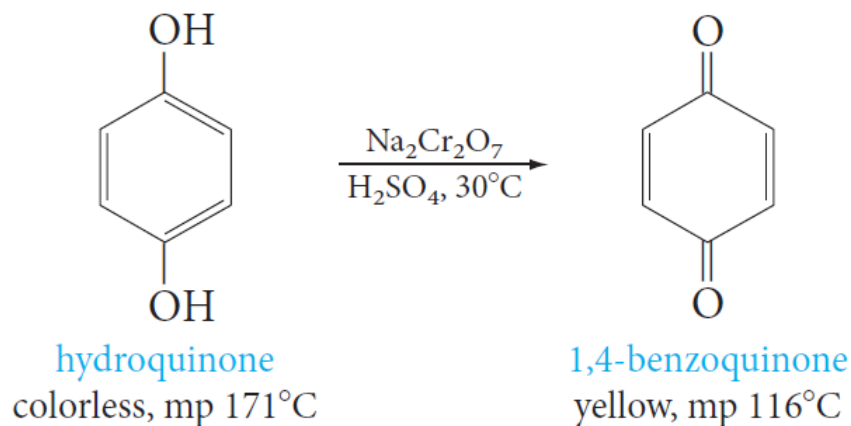
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PROBLEM 7.19 Write an equation for the oxidation of

- a. 4-methyl-1-octanol with Jones' reagent
- b. 4-methyl-1-octanol with PCC
- c. 4-phenyl-2-butanol with Jones' reagent
- d. 4-phenyl-2-butanol with PCC

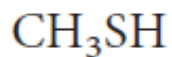
7.15 Oxidation of Phenols

- Reaction of a phenol with strong oxidizing agents yields a quinone

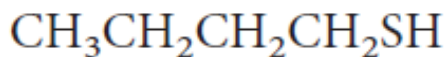


7.17 Thiols, Sulfur Analogs of Alcohols and Phenols

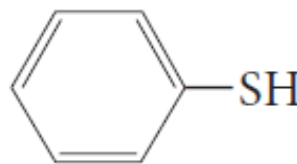
- **Thiols (RSH)**, are sulfur analogs of alcohols
 - Named with the suffix *-thiol*
 - SH group is called “mercaptan”



methanethiol
(methyl mercaptan)



1-butanethiol
(*n*-butyl mercaptan)



thiophenol
(phenyl mercaptan)

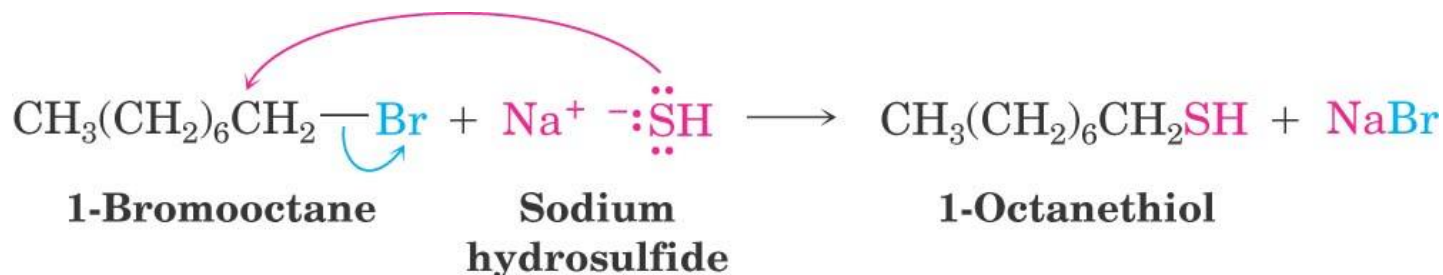
Thiols are more acidic than alcohols due to the size of sulfur S atom larger than O atom).

Thiols react with aqueous base to give thiolates.



Thiols: Formation and Reaction

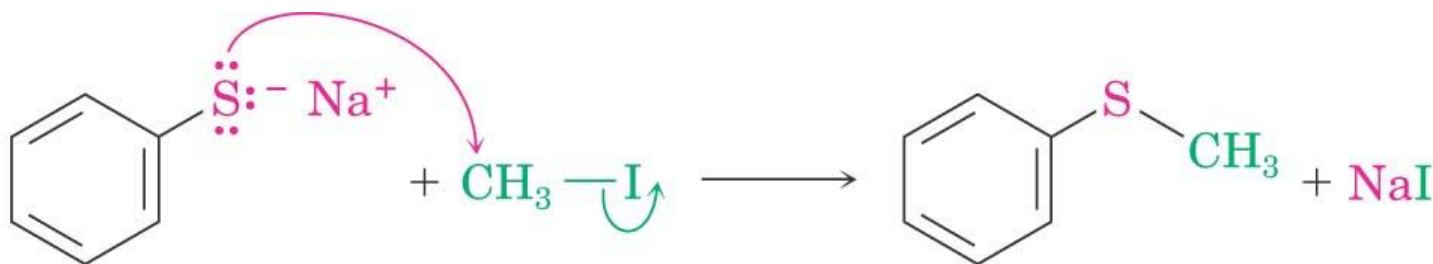
- From alkyl halides by displacement with a sulfur nucleophile such as -SH
 - The alkylthiol product can undergo further reaction with the alkyl halide to give a symmetrical sulfide, giving a poorer yield of the thiol



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Sulfides

- **Thiolates (RS^-)** are formed by the reaction of a thiol with a base
- Thiolates react with primary or secondary alkyl halide to give sulfides (RSR')
- Thiolates are excellent nucleophiles and react with many electrophiles

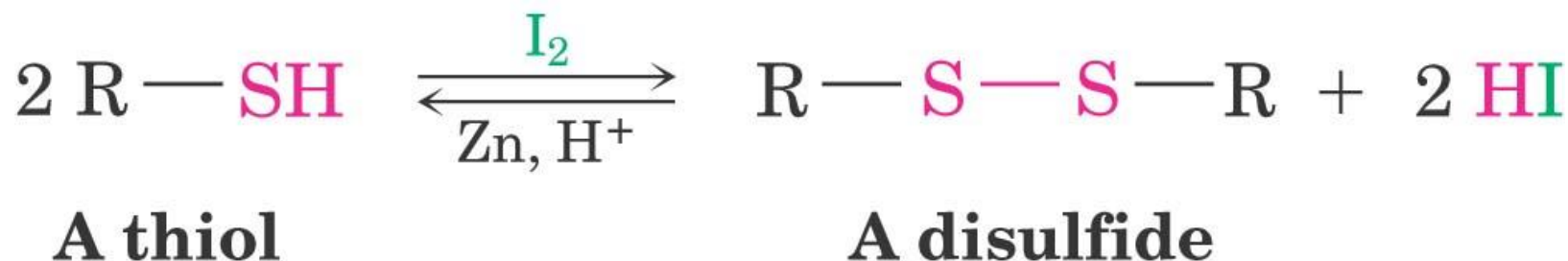


Sodium benzenethiolate

Methyl phenyl sulfide
(96%)

Oxidation of Thiols to Disulfides

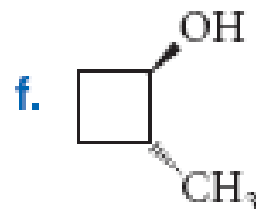
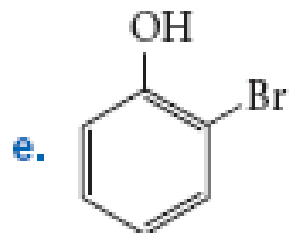
- Reaction of an alkyl thiol (RSH) with bromine or iodine gives a disulfide (RSSR)
- The thiol is oxidized in the process and the halogen is reduced



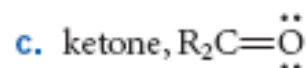
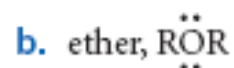
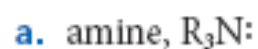
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The reaction can be reversed with a variety of reducing agents. Since proteins contain disulfide links, these reversible oxidation–reduction reactions can be used to manipulate the three-dimensional shapes of proteins.

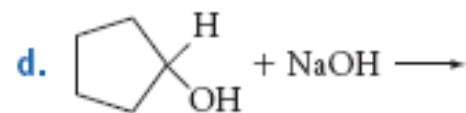
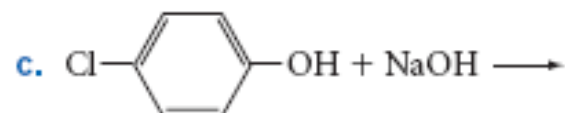
7.28 Name each of the following compounds:



7.35 The following classes of organic compounds are Lewis bases. Write an equation that shows how each class might react with H^+ .



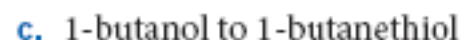
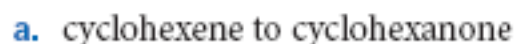
7.38 Complete each of the following equations:



7.48 Write an equation for each of the following reactions:



7.49 Write an equation for each of the following two-step syntheses:



7.54 Dimethyl disulfide, $\text{CH}_3\text{S}-\text{SCH}_3$, found in the vaginal secretions of female hamsters, acts as a sexual attractant for the male hamster. Write an equation for its synthesis from methanethiol.

7.29 Explain why each of the following names is unsatisfactory, and give a correct name:

a. 2-ethyl-1-propanol

b. 2,2-dimethyl-3-butanol

c. 1-propene-3-ol

d. 2-chloro-4-pentanol

e. 3,6-dibromophenol